

May 4, 2026

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Motivation

- ▶ Data compression is essential for modern computing.
- ▶ Lossless compression: Recovers original data exactly (e.g., LZSS, LZW).
- ▶ Need for high-speed decompression in data-intensive applications.

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Light Loss-Less Data Compression, With GPU Implementation

└ Introduction

└ Motivation

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- ▶ Lossless compression: Recovers original data exactly (e.g., LZSS, LZW).
- ▶ Need for high-speed decompression in data-intensive applications.

From today's perspective, data compression is crucial. We generate vast amounts of data daily, and chip prices are rising. Therefore, we need to reduce our data without losing any details.

At the same time, it is also important to retrieve the desired data from its compressed form.

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- Light Loss-Less Data Compression, With GPU Implementation
 - └ Introduction
 - └ Problem Statement

To reduce time and benefit from the massive parallel processing power of GPUs, we need a lossless compression system that can be parallelized.

- **Issue:** Dictionary-based methods update dictionaries sequentially.
- **Challenge:** Hard to parallelize for high-performance decompression on GPUs.
- **Goal:** Design a lossless compression method that allows massive parallelization on GPU.

- ▶ **LZSS/LZW:** Classic dictionary-based methods; hard to parallelize.
- ▶ **Huffman Coding/Arithmetic Coding:** Entropy-based; high compression but computationally expensive for GPUs.
- ▶ **Gzip/Zlib:** Industry standards; CPU-centric design.
- ▶ **Previous GPU Efforts:** Focus on specific data types or limited speedups due to sequential dependencies.

Briefly existing methods and their limitations in the context of GPU parallelization.

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Main Contribution

- ▶ **LLL (Light Loss-Less):** A new lossless compression method.
- ▶ Specifically designed for **high parallelization** on GPUs.
- ▶ Optimized for scenarios where decompression happens more frequently than compression (e.g., loading images or game assets).
- ▶ Achieves **91.1–176×** speedup over CPU implementation.

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Light Loss-Less Data Compression, With GPU Implementation

Existing Works / Literature Review

- └ Main Contribution

New compression method proposed by the authors.

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LLL Encoding Rules

Codes	words	length	encoded string
Non-dictionary encoding			
SC Single Character	c	1	c
RL Run-Length	c l	$l + 2$	$cc \cdots c$
Dictionary encoding			
SC Single Character	c	1	c
SI Short Interval (2-byte code)	t l	$l + 2$	$x(t) \cdots x(t + l + 1)$
LI Long Interval (3-byte code)	t 1111 c	$c + 18$	$x(t) \cdots x(t + c + 17)$
SRL Short Run-Length (2-byte code)	1111 1111 1111 l	$l + 2$	$pp \cdots p$
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Rules of codes: p is the previous character and $x(0)x(1) \cdots x(4095)$ is the previous segment.

Light Loss-Less Data Compression, With GPU Implementation

LLL Encoding

LLL Encoding Rules

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LLL Encoding Rules

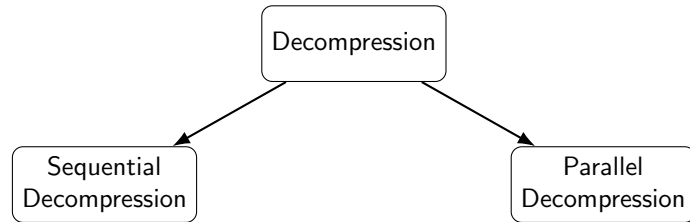
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Uses the previous segment (4096 bytes) as a dictionary.



Decompression Overview



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Light Loss-Less Data Compression, With GPU Implementation

- └ Algorithms
 - └ Decompression
 - └ Decompression Overview

Decompression Overview



This work has both sequential and parallel decompression technique.

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- Light Loss-Less Data Compression, With GPU Implementation
 - Algorithms
 - Sequential Decompression
 - Sequential Decompression Algorithm

...

- Reads word identifiers and byte/word arrays.
- For SC: Writes the character directly.
- For RL/Interval: Copies characters from dictionary or previous output.
- Complexity: $O(n)$ time.

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Lets see the sequential one first.

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This gives the complexity of $O(n)$.

Sequential decompression is $O(n)$, which is already fast, but we want even more speed via parallelization.



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Parallel Decompression on GPU

Two-step process using **Prefix-sums**:

- ▶ **Step 1: Calculate positions.**
 - ▶ Find where to write each code in the output.
 - ▶ Calculate where to read from and how much data for interval codes.
- ▶ **Stage 2: Perform copy operations.**
 - ▶ Each **thread** copies data to its assigned output location.
- ▶ **Work-Optimal:** Achieves $O(n)$ total work and $O(k \log m)$ time.

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Light Loss-Less Data Compression, With GPU Implementation

—Algorithms

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- Parallel Decompression on GPU

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Experiment Setup

- ▶ **GPU:** NVIDIA GeForce GTX 1080 (8GB VRAM).
- ▶ **CPU:** Intel Core i7-4790 (3.60 GHz).
- ▶ **Software:** CUDA Toolkit, C++ Compiler.
- ▶ **Dataset:** Standard benchmark images and data (e.g., Crafts, Flowers, Graph, Random, Black).

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Experiment Setup

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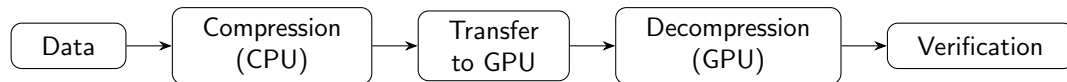
This work was conducted in 2016, when the NVIDIA GeForce GTX 1080 was the flagship GPU used in the experiments.

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Experiment Steps



1. Compress input data using LLL on CPU.
2. Transfer compressed data from Host to Device.
3. Measure decompression time on GPU for different datasets.
4. Compare with sequential CPU decompression and other methods (LZSS, LZW).

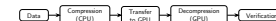
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So, how this experiment was conducted?.

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- Light Loss-Less Data Compression, With GPU Implementation
 - Experimental Results
 - Performance Evaluation

The authors found that ...



Answering Research Question 1

RQ1: Can LLL achieve significant speedup on GPU while maintaining competitive compression ratios?

- ▶ **Yes.** LLL achieves $91\text{--}176\times$ speedup over CPU.
- ▶ Compression ratio is comparable to standard LZSS/LZW for the tested datasets.

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Light Loss-Less Data Compression, With GPU Implementation

└ Answering Research Questions

└ Answering Research Question 1

Here comes our research questions. . .

The compression ration was similar to the CPU implementations, but the decompression speed was significantly higher on GPU.

Answering Research Question 1

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Answering Research Question 2

RQ2: How does LLL compare to existing parallel-friendly methods?

- ▶ LLL outperforms LZW (2.49–9.13 $\times$) and LZSS (4.30–14.1 $\times$) in decompression speed on GPU.
- ▶ Its prefix-sum based approach is more work-efficient.

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- └ Discussion / implication

Discussion / implication

- ▶ **Real-time Applications:** Ideal for asset loading and real-time texture decompression.
- ▶ **Architecture Synergy:** LLL's design aligns perfectly with GPU warp execution patterns.
- ▶ **Data Transfer Bottleneck:** Highlights that GPU decompression can alleviate the PCIe bottleneck by transferring less data.

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Threats to Validity

- ▶ **Internal:** Prefix-sum overhead may be more noticeable on very small datasets; experiments use grayscale samples.
- ▶ **External:** Results might vary on different GPU architectures (e.g., AMD vs. NVIDIA).
- ▶ **Construct:** Performance depends on the "repetitiveness" of data (common in images but less so in random binary).

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- ▶ Massive acceleration achieved on GPU compared to standard methods.
- ▶ Future work could include multi-GPU (cluster) support or hybrid compression schemes.

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Conclusion

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LLL decompression on the GPU is much faster than previously published LZW decompression and LZSS decompression.

This shows that GPU LLL decom-pression can be useful for many applications.

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Thank You!
Questions?

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Thank you very much for your time and attention